Networking and Technical Support

AITS Learning Hub training manual and practice workbook

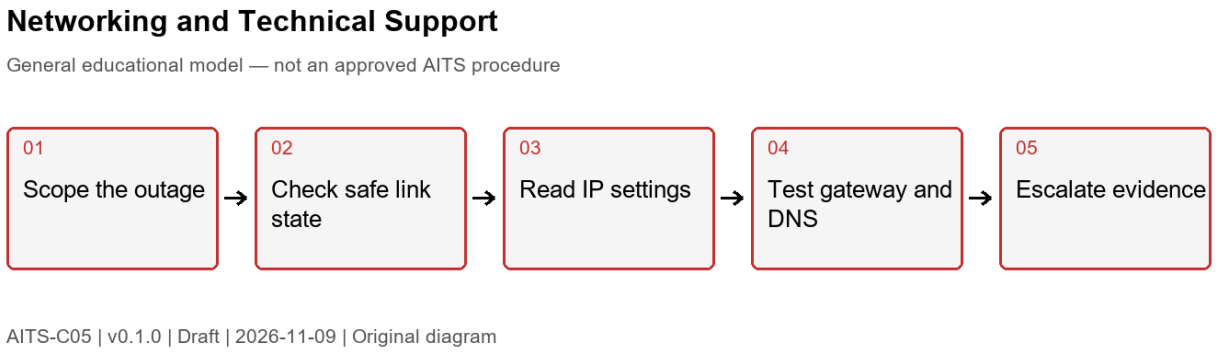
Sample Content — Pending Company Approval

This course introduces the concepts below and gives you exercises to practise them. Use the source notes to distinguish public company information, general guidance and fictional examples. It does not authorize live work or establish internal policy.

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Language	English; Thai summaries pending native-speaker review

Learning objectives

- Explain the path from a device to an external service.
- Distinguish an address, a gateway and name resolution.
- Choose diagnostic steps that test a hypothesis.
- Hand over an investigation with evidence and a clear next action.



LAN switches routers and Wi-Fi

AITS-C05-L01 | 15 minutes | Draft

Objective: Explain the path from a device to an external service.

Prerequisites: AITS-C01

A local area network connects nearby devices. A switch commonly connects wired devices within that network, a router forwards traffic between networks, and an access point gives wireless devices access to a network. One physical product may combine these functions. A Wi-Fi connection alone does not prove internet access.

Trace the path from the client through access, local switching, gateway and upstream connection to the service. A fault can affect one device, one wireless area, one VLAN or the whole site. Start by asking what works as well as what fails. Comparing affected and unaffected paths narrows the investigation without changing the network.

Worked example

A fictional laptop sees the office Wi-Fi name but cannot open websites. Another wired workstation works. Begin with the laptop/wireless path rather than declaring a site-wide ISP outage.

Try it

Draw a laptop-to-internet path and mark which observations would distinguish a client fault from a shared fault.

Your response: _____

Compare your reasoning

Laptop → access point → switch → router → provider → service. Compare another client, wired access and another service using authorized tests.

Remember

Wi-Fi connection is not proof of internet Separate device and shared scope Compare working and failing paths

Thai summary (AI translation, not professionally reviewed): การเชื่อมต่อ Wi-Fi ไม่ยืนยันว่าใช้อินเทอร์เน็ตได้ เปรียบเทียบเส้นทางที่ใช้งานได้กับเส้นทางที่มีปัญหา

References: NET, EX. See source register.

IP addressing DHCP and DNS

AITS-C05-L02 | 15 minutes | Draft

Objective: Distinguish an address, a gateway and name resolution.

Prerequisites: AITS-C05-L01

An IP address identifies an interface on a network; the subnet prefix helps determine which destinations are local. A default gateway provides a path to other networks. Private IPv4 ranges include 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16; they are not globally routed public addresses. Avoid duplicate assignments and do not change addressing without an approved plan.

DHCP supplies configuration such as an address lease and network options. DNS resolves names to records used to locate services. These functions are different: a valid lease does not prove DNS or upstream access works. In the synthetic 192.168.10.0/24 exercise, a client at .20 and gateway at .1 share the subnet; .0 and .255 are not ordinary host choices for that subnet.

Worked example

A lab client has 192.168.10.20/24, gateway 192.168.10.1 and an approved resolver. Record all three before drawing a conclusion about a failed website.

Try it

Explain the difference between DHCP and DNS to a nontechnical colleague.

Your response: _____

Compare your reasoning

DHCP helps a device obtain network configuration; DNS helps translate a service name into the records needed to find it.

Remember

Address, prefix and gateway work together DHCP supplies configuration DNS handles names

Thai summary (AI translation, not professionally reviewed): IP ระบุอินเทอร์เฟซ DHCP แจกค่าตั้งเครือข่าย และ DNS ช่วยค้นหาปลายทางจากชื่อ ทั้งสามหน้าที่แตกต่างกัน

References: IP, DHCP, NET, EX. See source register.

Investigate internet unavailable

AITS-C05-L03 | 15 minutes | Draft

Objective: Choose diagnostic steps that test a hypothesis.

Prerequisites: AITS-C05-L02

First establish safety, authorization, affected users, services, start time and recent changes. Ask whether all sites fail or one application fails. Observe link and connection state safely. Read the client's address, prefix, gateway and DNS configuration before changing it. Compare a known-good device on the relevant network.

On an authorized test device, basic diagnostics can include ipconfig /all, a gateway reachability test and nslookup for an approved test name. A ping timeout is evidence, not proof that a device is offline: filtering or policy may block replies. If gateway and external connectivity appear available but name resolution fails, investigate the configured resolver and related policy. Change one authorized variable at a time, record before/after results and preserve a rollback route.

Worked example

Synthetic outage: all clients reach the gateway; an approved external TCP test succeeds; lookups through the configured resolver time out. This supports a DNS-path hypothesis, not a confirmed broken cable.

Try it

Choose the next two read-only checks and explain what each result would tell you.

Your response: _____

Compare your reasoning

Compare resolver configuration with the approved baseline and test the configured resolver from another affected client. Record results and escalate before changing settings.

Remember

Define scope and recent changes Observe before changing A failed ping is not a complete diagnosis

Thai summary (AI translation, not professionally reviewed): กำหนดขอบเขตปัญหาและเก็บข้อมูลก่อนเปลี่ยนค่า Ping ไม่ตอบไม่ได้แปลว่าอุปกรณ์ปิดเสมอไป

References: NET, DHCP, EX. See source register.

Isolation escalation and customer updates

AITS-C05-L04 | 15 minutes | Draft

Objective: Hand over an investigation with evidence and a clear next action.

Prerequisites: AITS-C05-L03

Device isolation has two meanings: narrowing a fault to a device/path, and restricting a device for security containment. Do not disconnect equipment or change VLANs under the second meaning without the applicable authorization. For diagnosis, compare a known-good cable, port or client only where the test is safe and approved. Preserve the original state and record each change.

An escalation includes impact, scope, timestamps, relevant topology, tests and results, changes made, rollback status and the requested help. Exclude passwords and unnecessary personal information. Tell the customer what is known, what remains uncertain and when the next agreed update will occur. Do not promise resolution times or quote an AITS SLA that has not been verified.

Worked example

Synthetic update: “Two office devices cannot resolve names. Gateway checks passed. We are checking the resolver path and will update you at the agreed time.”

Try it

Write a support escalation with three observations and one explicit request.

Your response: _____

Compare your reasoning

Two clients affected since 09:10; gateway reachable; configured resolver times out. No configuration changed. Please review resolver/service policy and advise the authorized next test.

Remember

One controlled comparison at a time Escalate evidence and uncertainty Promise only an agreed update

Thai summary (AI translation, not professionally reviewed): ส่งต่อผลทดสอบ การเปลี่ยนแปลง และสิ่งที่ยังไม่ทราบ แจ้งเวลาติดตามที่ตกลงกันโดยไม่รับรอง SLA ที่ยังไม่ยืนยัน

References: EX, QA. See source register.

Practical scenarios

Internet unavailable

Synthetic classroom exercise; not an AITS procedure. Role: Support trainee

Lotus Workshop reports no internet on three devices. Gateway responds; approved external TCP test passes; configured DNS lookups time out.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Check resolver configuration and compare another client, then escalate evidence.

Inappropriate: Factory-reset the router before recording settings.

Consequence: A focused DNS investigation preserves the working network path.

Only one laptop fails

Synthetic classroom exercise; not an AITS procedure. Role: Helpdesk trainee

One fictional laptop cannot browse; nearby clients work. The affected laptop has an unexpected gateway in the lab snapshot.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Compare with the approved baseline and request authorized correction.

Inappropriate: Change every client address.

Consequence: A scoped correction avoids expanding a single-device fault.

Ping is blocked

Synthetic classroom exercise; not an AITS procedure. Role: Technician trainee

A device does not reply to ping but its authorized web interface opens. ICMP filtering is possible; application access succeeds.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Record both results and test the actual service requirement.

Inappropriate: Declare the device dead from ping alone.

Consequence: Multiple observations prevent a false diagnosis.

Use the full scenario JSON for the four-criterion rubric. Score evidence, decision quality, authority/safety and handover from 0–2 each. 0 = absent or unsafe; 1 = partial; 2 = complete and supported. Scores are formative, not certification.

Knowledge check

1. What commonly connects different IP networks?

A. Passive label | B. Cable tray | C. Router

2. Does a Wi-Fi icon prove a website is reachable?

A. Only at full signal | B. No | C. Yes

3. One laptop fails while a wired workstation works. What is useful next?

A. Compare the affected access path | B. Reset the entire site | C. Declare every server down

4. Which is a private IPv4 example?

A. A MAC label | B. 192.168.10.20 | C. A domain name

5. Which function commonly leases client addressing?

A. DHCP | B. DNS | C. A document editor

6. What does DNS primarily help with?

A. Supplying PoE power | B. Replacing a damaged cable | C. Resolving names to records

Use the answer key after attempting the questions. Proposed practice pass mark: 80%. This is not an AITS competency rule.

Knowledge check continued

7. What should come before resetting the router?

A. Scope, evidence and authorization | B. Repeated resets | C. Replacing every cable

8. Does a ping timeout prove a device is off?

A. Yes | B. Only on Wi-Fi | C. No

9. Gateway and external service tests work but name lookup fails. What is a reasonable hypothesis?

A. A proven mains fault | B. A DNS-path issue | C. A guaranteed camera fault

10. What belongs in an escalation?

A. Customer passwords | B. Only “internet broken” | C. Tests, results and changes made

11. Should you disconnect a critical device without authorization?

A. Only if an AI suggests it | B. No | C. Yes for every fault

12. A ping timeout alone proves a device is powered off.

A. True | B. False

Use the answer key after attempting the questions. Proposed practice pass mark: 80%. This is not an AITS competency rule.

Answer key and references

1. C — Routing forwards traffic between networks.
2. B — Wireless association is only one part of the path.
3. A — Comparison narrows the scope without disruption.
4. B — 192.168.0.0/16 is a private-address block.
5. A — DHCP distributes network configuration.
6. C — Name resolution is distinct from physical connectivity.
7. A — A reset may disrupt users and erase useful evidence.
8. C — Filtering or policy can prevent replies.
9. B — The evidence narrows the next checks but is not final proof.
10. C — Specific evidence lets the next person continue.
11. B — Containment and service impact require the approved authority.
12. B — A timeout can reflect filtering or policy; compare other authorized service tests before deciding that a device is off.

DHCP: RFC 2131 DHCP. 1997-03; accessed 2026-11-09. https://www.rfc-editor.org/rfc/rfc2131

EX: Original instructional examples. 2026-11-09. training-content/build_content.py

IP: RFC 1918 private addressing. 1996-02; accessed 2026-11-09. https://www.rfc-editor.org/rfc/rfc1918

NET: Cisco networking basics. Accessed 2026-11-09. https://www.cisco.com/site/us/en/learn/topics/small-business/networking-basics.html

QA: ISO quality assurance overview. Accessed 2026-11-09. https://www.iso.org/quality-management/quality-assurance